



10. Which Java Card API method is used to send a response in an APDU transaction?
 A) apdu.setOutgoing(); B) apdu.sendBytes();
 C) apdu.sendAPDU(); D) apdu.writeResponse();
11. What is the main function of Java Card APIs?
 A) To facilitate interaction between Java Card applications and the underlying smart card platform
 B) To provide a graphical user interface for Java applications
 C) To improve the execution speed of Java applications
 D) To replace the need for a Java Card Virtual Machine
12. What is the correct way to start a transaction in Java Card?
 A) JCSYSTEM.beginTransaction(); B) JCSYSTEM.startTransaction();
 C) Transaction.begin(); D) startTransaction();
13. Which of the following is the correct order of the Java Card applet development process?
 A) Writing Code → Compilation → Deployment → Testing
 B) Compilation → Writing Code → Testing → Deployment
 C) Testing → Compilation → Writing Code → Deployment
 D) Deployment → Compilation → Writing Code → Testing
14. How do you commit an atomic transaction in Java Card?
 A) JCSYSTEM.endTransaction(); B) JCSYSTEM.completeTransaction();
 C) JCSYSTEM.finishTransaction(); D) JCSYSTEM.commitTransaction();
15. How does Java Card manage memory for objects?
 A) By using an automatic garbage collector
 B) By manually deallocating memory after use
 C) By allocating objects in EEPROM or RAM
 D) By using a reference counting mechanism depending on their use
16. What is the correct way to define a persistent byte array in Java Card?
 A) byte[] buffer = new byte[256];
 B) byte[] buffer = JCSYSTEM.makeTransientByteArray(256, JCSYSTEM.CLEAR_ON_RESET);
 C) byte[] buffer = JCSYSTEM.makePersistentByteArray(256);
 D) byte[] buffer = new byte[256]();
17. Which method is called to register a Java Card applet instance during installation?
 A) install() B) start()
 C) init() D) register()
18. How do you retrieve the APDU buffer from an APDU object?
 A) apdu.getBuffer(); B) apdu.getDataBuffer();
 C) apdu.fetchBuffer(); D) apdu.receiveBytes();
19. Which exception should be thrown if an unsupported APDU instruction is received?
 A) APDUException.ILLEGAL_USE B) ISOException.throwIt(ISO7816.SW_INS_NOT_SUPPORTED);
 C) SystemException.throwIt(SystemException.ILLEGAL_VALUE); D) SecurityException.throwIt(SecurityException.UNAUTHORIZED);
20. Which tool is commonly used for Java Card applet testing?
 A) NetBeans IDE B) Eclipse Java EE
 C) GlobalPlatform Pro D) Android Studio

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the key differences between embedded system security and traditional IT security. Why is embedded security more challenging?	8	L*2	1	1
	b) What are the key steps in a secure development process for embedded systems?	8	L2	2	1
2.	a) Discuss the different types of hardware and software-based attacks on embedded systems. Provide suitable mitigation strategies.	8	L2	1	1

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|-------|--|---|----|---|---|
| b) | Explain the role of hypervisors in embedded system security. What are their advantages? | 8 | L2 | 2 | 1 |
| 3. a) | What are security policies in embedded systems? Explain different types with examples. | 8 | L2 | 1 | 1 |
| b) | How does model-driven design improve security in embedded systems? Provide a case study example. | 8 | L2 | 2 | 1 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Explain the working of Java Card Virtual Machine (JCVM) and its execution model. | 8 | L2 | 3 | 1 |
| b) | Explain the challenges in the development of smart card applications. | 8 | L2 | 4 | 1 |
| 5. a) | Explain the significance of the Hardware Abstraction Layer (HAL) in the smart card software architecture. | 8 | L2 | 3 | 1 |
| b) | Explain the lifecycle of a Java Card applet with an example. | 8 | L2 | 4 | 1 |
| 6. a) | Describe the architecture and responsibilities of the Java Card Runtime Environment (JCRE). | 8 | L2 | 3 | 1 |
| b) | What is atomicity in Java Card transactions? Explain with an example. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Explain the following: | | | | |
| | i. Commit transaction | 8 | L2 | 5 | 1 |
| | ii. Abort transaction | | | | |
| b) | Describe the exception handling mechanism in Java Card and its importance. | 8 | L2 | 5 | 1 |
| 8. a) | What are APDU commands in Java Card, and how are they processed? | 8 | L2 | 5 | 1 |
| b) | Explain the following: | | | | |
| | i. ISOException | 8 | L2 | 5 | 1 |
| | ii. Java Card Exception Reason Code | | | | |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
